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NODE 2 — INTRAVAGINAL LONGITUDINAL PHYSIOLOGY MONITOR FOR EXTENDED WEAR

A. Overview

In certain embodiments, a second node is configured as an intravaginal longitudinal physiology monitor intended for extended wear to measure tissue-adjacent physiologic patterns during everyday conditions. The second node may operate for hours at a time and may collect signals suitable for longitudinal pattern measurement, within-user trend comparison, and cross-device correlation with one or more other nodes.

B. Form Factor and Wear Geometry

The second node may be implemented as an intravaginal insert having a soft external body and an internal support structure. The body may include an insertion geometry configured for comfortable placement and stable positioning, including one or more of tapered ends, an hourglass profile, compliant flanges, flexible collars, textured microstructures, or distributed compliance regions. The housing may include sealing features for fluid resistance and cleaning, and may include a retrieval feature such as a flexible tail, loop, tab, or magnetic retrieval accessory. In certain embodiments, the device includes multiple sensing zones distributed along an axial length to capture spatially distinct signals.

C. Sensor Suite

The second node may include one or more sensors and sensing elements, including combinations of the following: 1. Bio-impedance sensing, including multi-frequency impedance sensing, to characterize tissue-adjacent electrical properties over time. 2. Capacitive wetness/dielectric sensing to characterize moisture state, dielectric changes, and related patterns. 3. Solid-state pH sensing, including ISFET-based pH sensing, to characterize pH dynamics over time. 4. Tissue hydration sensing, including impedance- or dielectric-based hydration proxies. 5. Temperature sensing, including one or more temperature sensors to provide thermal context and support wear detection and artifact rejection. 6. Pelvic activity sensing, including detection of muscular activity via pressure dynamics, impedance dynamics, and/or other suitable signals. 7. Distributed pressure sensing, including capacitive pressure, piezoresistive, and/or FSR-based pressure sensing, optionally in multiple zones. 8. MEMS sensing, including one or more MEMS sensors used for pressure, motion, or environmental sensing. 9. Motion/orientation sensing, including 6-axis IMU sensing, for artifact rejection, motion gating, and robustness. Sensors may be arranged circumferentially, axially, or in segmented zones. In some embodiments, electrodes for impedance and/or capacitive sensing are configured as rings, arcs, pads, bands, interdigitated electrodes, or other patterns.

D. Data Collection, Wear Detection, and Artifact Rejection

The second node may operate in continuous sampling mode, periodic sampling mode, or event-triggered mode. Wear detection may be performed using one or more of impedance baselines, capacitive proximity/contact confirmation, temperature changes, pressure baselines, and/or multi-sensor agreement. The system may compute a signal quality index to identify invalid segments due to motion, insertion misalignment, poor contact, sensor saturation, or environmental changes. Artifact rejection may include inertial-based gating, contact-integrity checks, cross-sensor consistency checks, and exclusion of low-confidence segments. In certain embodiments, the system identifies stable measurement windows based on low-motion states and consistent contact measures, and tags segments with confidence indicators.

E. Derived Features and Metrics

The computing system may derive features from Node 2 signals including impedance spectra features, temporal trends, cyclic patterns, variability measures, event detection features, pelvic activity signatures, hydration/wetness signatures, pH signatures, and temperature-context features. Metrics may be expressed using privacy-preserving abstractions, including indices, zones, ranges, and curves. Example metrics include: a hydration or wetness index derived from dielectric/capacitive measures, an impedance-based tissue-state index derived from multi-frequency impedance features, a pH trend curve derived from solid-state pH sensing, a pelvic activity signature derived from pressure and motion-conditioned features, and a confidence indicator representing segment validity and contact stability.

F. Cross-Device Correlation and Context Conditioning

Node 2 outputs may be correlated with outputs from an upper-body context node to condition interpretation based on activity state, stress proxies, temperature context, sleep context, or respiratory dynamics. Node 2 outputs may also be correlated with baseline acquisition results and/or internal geometry/compliance metrics from other nodes to improve normalization, interpretability, and cross-session comparability.

G. Variations and Alternatives

The second node may be implemented with different combinations of sensors and electrode patterns. Certain embodiments emphasize impedance and dielectric sensing; other embodiments emphasize pH sensing; other embodiments emphasize distributed pressure sensing. The node may include local storage, intermittent upload, or real-time streaming. Power may be provided via a rechargeable battery, inductive charging, or other power subsystems. The housing may be fully sealed and formed using medical-grade elastomers with overmolding, multi-material structures, or modular sensor inserts.